Proposed Project Extensions & Innovations

SENG 495 - TA3. Chemotherapy Response and Tumor Suppression

To build upon the baseline PINN methodology, we have brainstormed several potential directions ranging from mathematical extensions to software engineering platforms. We would like to discuss which of these paths aligns best with the course objectives:

HEALTHTECH PRODUCT

Patient-Specific Treatment Sandbox (Forward Simulation)

The baseline study estimates static parameters but lacks predictive interaction. We propose developing a "sandbox" simulation module. Once the PINN discovers a specific patient's cell-kill coefficients, doctors can input hypothetical future treatment schedules (varying dosage or frequency) to simulate and visualize how the tumor will dynamically respond, creating a personalized treatment planner.

HEALTHTECH PRODUCT

Role-Based Clinical Web Application

Transforming the mathematical core into a deployable healthcare application. We will build a web platform with two distinct interfaces: A **Doctor Portal** to upload measurements, run the PINN, and simulate treatments, and a simplified **Patient Portal** where patients can securely log in to view their tumor regression progress and understand their treatment trajectory via clear dashboards.

SOFTWARE ENGINEERING & TESTING

End-to-End Tumor Response & Consistency Platform

Developing a complete software tool featuring automated metamorphic testing. Beyond predictions, the software will verify physical consistency (e.g., ensuring outputs remain invariant whether time is inputted in days or hours). It will include a classical numerical ODE solver to benchmark against the PINN, allowing us to deliberately inject synthetic errors to measure the system's fault-detection capabilities.

CLINICAL RELEVANCE

Early-Stage Prediction Analysis

The baseline study monitors measurements over a full 105-day period. In clinical settings, waiting 100 days to evaluate treatment efficacy is impractical. We will truncate the dataset to the first 20, 30, and 50 days to test whether the PINN can still accurately infer the underlying tumor growth and drug-sensitivity parameters early enough for timely medical intervention.

MATHEMATICAL MODEL

Integration of a Drug Resistance Parameter

Real-world tumors develop resistance to chemotherapy over time. We can modify the underlying ODE system to include a time-decaying sensitivity parameter, allowing the PINN to predict dynamic, rather than static, cell-kill coefficients as the treatment progresses.

MATHEMATICAL MODEL

Multi-Drug Dynamics (Combination Therapy)

Extending the existing single-drug compartmental model to accommodate two synergistic interventions (e.g., dual chemotherapy or concurrent radiotherapy) and estimating their combined parameters using the physics-informed architecture.

DATA INTEGRATION

Validation via Open-Source Clinical Data

Replacing or augmenting the synthetic baseline data with real patient tumor-volume datasets from platforms like *The Cancer Imaging Archive (TCIA)* to test the model's robustness against noisy, real-world clinical data.

ARCHITECTURE IMPROVEMENT

Self-Adaptive PINN Architecture

Upgrading the neural network architecture from a standard approach to a Self-Adaptive PINN, where the weighting of the data-driven and physics-informed loss terms are dynamically updated during training to improve convergence speed and accuracy.